Bhumika Mittal

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Research Overview

I work at the intersection of programming languages and cryptography. My current research focuses on equivalence checking and code optimization for finite-field computations, particularly in cryptographic implementations.

Education

2026–present **Georgia Institute of Technology**
Ph.D., Computer Science
Advisor: Qirun Zhang

2021–2025 **Ashoka University**
Postgraduate Diploma in Advanced Studies and Research
Thesis: *Ring Trapdoor Functions: A Lattice-Based Framework for Secure Ring Signatures*

B.Sc. (Honours), Computer Science
Minors in Mathematics and Entrepreneurial Leadership & Strategy

Publications

The Trust a Protocol Needs
with Aalok Thakkar
International Conference on Information Systems Security (ICISS 2026)

Defining Purpose-Limited Secrets
with Aalok Thakkar
International Conference on Information Systems Security (ICISS 2026)

AxLaM: Energy-Efficient Accelerator Design for Language Models for Edge Computing
with Tom Glint, Santripta Sharma, Abdul Ronak, Abhinav Goud, Neerja Kasture, Zaqi Momin, Aravind Krishna, and Joycee Mekie
Philosophical Transactions of the Royal Society A (2025)

On the Existence of Balanced Generalized de Bruijn Sequences
with Matthew Baker, Haran Mouli, and Eric Tang
Discrete Mathematics (2023)

Research Experience

2024 **Max Planck Institute for Software Systems**
Visiting Research Fellow
Built a versioned dynamic graph store with 72% lower read latency and 31% higher throughput than lock-based designs, and designed PACT, a phase-concurrency metric.

2024 IIT Delhi
Research Assistant
Designed a reduction from the Closest Vector Problem to weighted MaxSAT for even norms, and evaluated its scalability on the RC2 MaxSAT solver.

2023–2024 Ashoka University
Research Intern
Designed and analyzed lattice-based schemes for the quantum-safe migration of communication systems, and implemented a constant-time cryptographic module that is now in active use.

2023 IIT Gandhinagar
Research Assistant
Co-designed AxLaM, a language-model accelerator with 9× lower energy and 58% smaller area than prior designs.

Industry Experience

2025–2026 Amuse Labs, Bengaluru, India
Software Engineer
Added validation and error handling across content-import and editing workflows of the PuzzleMe platform, extended its end-to-end test suite, and built accessibility support.

2021–2022 WebVeda
Tech and Product Manager
Developed an online learning platform that handled high traffic and generated substantial revenue within its first year, and managed its end-to-end pipeline.

Honors and Awards

2025 Academic Excellence Award, Department of Computer Science, Ashoka University

2025 Teaching Assistantship Excellence Award, Department of Computer Science, Ashoka University

2024 Summa Cum Laude, Ashoka University

2024 Silver Medal, Department of Computer Science, Ashoka University

2024 Builder's Award for Service Excellence, Department of Computer Science, Ashoka University

2022 Undergraduate Research Excellence Award, Department of Mathematics, Ashoka University

2021–2025 Deutsche Bank Scholarship, Ashoka University

2020–2021 Mehta Fellow, Program in Mathematics for Young Scientists (PROMYS), Boston University

Travel Grants

2025 Mentoring Workshop and Conference, FSTTCS 2025, Goa

2025 Programming Languages Mentoring Workshop, PLDI 2025, Seoul

2024	Mathematics for Post-Quantum Cryptanalysis, Budapest
2024	Undergraduate Architecture Mentoring Workshop, ISCA 2024, Buenos Aires

Teaching

2026	Georgia Institute of Technology *Teaching Assistant*: Programming Languages
2022–2025	Ashoka University *Teaching Assistant* for seven courses, including Information Security, Data Structures, Discrete Mathematics, and Introduction to Computer Programming (average student feedback 4.88/5.00)
2023	Summer Programs *Instructor*: Microcontrollers, Game Design, and Robotics at Young Tech Scholars, Plaksha University, and the Young Scholars Programme, Ashoka University *Teaching Assistant*: Science Communication, Lodha Genius Program

Service

Reviewing	Subreviewer: HPCA 2027, OOPSLA 2026, PLDI 2025 Student Research Competition
Organizing	Co-organizer, Workshop on Lattice-based Post-Quantum Cryptography, Ashoka University, 2024

Student Leadership

2024–2025	Finance Head, Quizzing Club, Ashoka University
2023–2024	Computer Science Student Representative, Academic Affairs Board, Ashoka University
2023–2024	Director of Technology, IEEE Ashoka Student Branch
2022–2024	Interim President and Student Advisor, Computer Science Society, Ashoka University

Technical Skills

Languages	OCaml, TypeScript, Java, C/C++, x86 and MIPS assembly
Verification	Lean 4, Z3, MaxSAT solvers
Areas	Program equivalence, code optimization, logic in computer science, finite-field arithmetic, lattice-based cryptography